Cloud Computing

What is Cloud Computing?

Cloud Computing is a technology that allows users to access computing resources over the internet instead of using local machines or physical servers.

❖ Key Idea:

You don’t own infrastructure → you **rent it on demand**.

❖ Services Provided:

- Computing Power (servers)
- Storage (databases, file systems)
- Networking
- Software applications
- AI/ML tools

❖ Real-Life Examples:

- Google Drive → store files online
- Gmail → email without installing software
- Netflix → streaming without downloading

Key Characteristics of Cloud Computing

1. **On-Demand Self-Service**
   Users can create resources instantly without human interaction.
2. **Broad Network Access**
   Accessible from mobile, laptop, tablet via internet.
3. **Resource Pooling**
   Multiple users share the same infrastructure securely.
4. **Rapid Elasticity (Scalability)**
   Scale up/down instantly based on demand.
5. **Measured Service (Pay-as-you-go)**
   You are charged based on usage.

Microsoft Azure Cloud

What is Microsoft Azure Cloud?

Microsoft Azure (Microsoft Azure) is a **cloud platform** that provides 200+ services for building and managing applications.

❖ Core Services in Azure

1. Compute Services

- Virtual Machines (VMs)
- Containers (Docker, Kubernetes)

2. Storage Services

- Blob Storage (files, images, videos)
- Disk Storage
- Queue Storage

3. Networking

- Virtual Networks (VNet)
- Load Balancers
- VPN Gateway

4. Databases

- SQL Database
- NoSQL (Cosmos DB)

5. AI & Machine Learning

- Pre-built AI models
- Custom ML training

6. DevOps Tools

- CI/CD pipelines
- Automation tools

❖ Why Companies Use Azure?

- Global data centers
- High security standards
- Integration with Microsoft tools (Windows, Office)

❖ Other Cloud Providers

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)

Advantages of Cloud Computing

1. Cost Savings

- No hardware purchase
- No maintenance cost
- Reduced IT staff requirement

2. Scalability & Flexibility

- Increase resources during high traffic
- Decrease when not needed

3. High Performance

- Latest hardware and infrastructure
- Faster processing and response time

4. Reliability

- Data backup across multiple servers
- Failover systems reduce downtime

5. Security

- Data encryption
- Identity and access management
- Firewalls and monitoring

6. Disaster Recovery

- Quick recovery from data loss or failure
- Backup stored in different regions

7. Mobility (Work from Anywhere)

- Access data anytime, anywhere
- Supports remote work culture

8. Automatic Software Updates

- No manual updates required
- Always use latest version

9. Collaboration Efficiency

- Multiple users can work on same file
- Real-time updates

10. Environment Friendly

- Reduces physical hardware usage
- Saves energy

Use Cases of Cloud Computing

1. Data Storage & Backup

- Store large data securely
- Automatic backup and restore

Example: Google Drive, OneDrive

2. Website & Application Hosting

- Host websites without buying servers
- Supports global users

Example:

- E-commerce sites
- Blogging platforms

3. Big Data Analytics

- Analyze huge datasets
- Used in:
  - Banking
  - Healthcare
  - Marketing

4. Artificial Intelligence & Machine Learning

- Train models using cloud power
- No need for high-end systems

Example:

- Chatbots
- Face recognition

5. Disaster Recovery & Backup Systems

- Recover data after cyber attacks or crashes
- Business continuity

Types of Cloud Deployment Models

1. Public Cloud

- Available to everyone
- Managed by providers like Azure

2. Private Cloud

- Used by a single organization
- More secure but costly

3. Hybrid Cloud

- Combination of public + private cloud
- Flexible and secure

Types of Cloud Services

1. IaaS (Infrastructure as a Service)

- Provides virtual servers and storage
- User controls OS and applications

Example: Azure Virtual Machines

2. PaaS (Platform as a Service)

- Provides platform to build apps
- No need to manage hardware

Example: Azure App Service

3. SaaS (Software as a Service)

- Ready-to-use applications

Example:

- Gmail
- Microsoft 365